AEMO's Engineering Framework & Technology Roadmap

Preparing for a high renewable future

IRED - 25 October 2022

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Modelling & Engineering



We acknowledge the Traditional Owners of country throughout Australia and recognise their continuing connection to land, waters and culture.

We pay respect to their Elders past, present and emerging.

The challenge facing us





- Old tech
- Manual



- Adding new technology – in flight!





- Power system of the future
- Automated / intelligent controls

PLUS
- BYO propulsion!
- Bigger plane (double by 2050!)

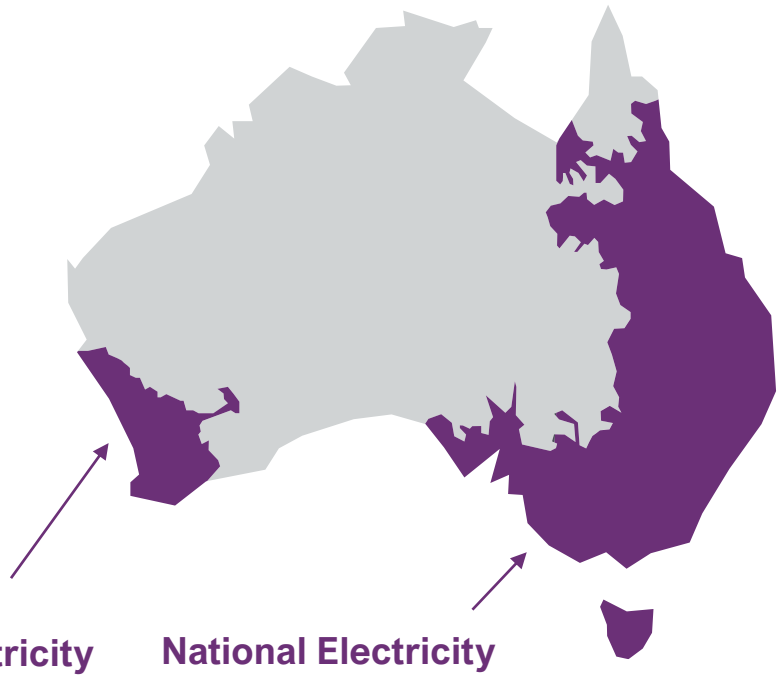
The Australian Energy Market Operator (AEMO)

AEMO is a member-based, not-for-profit organisation.

We are the **independent energy market and system operator and system planner** for the National Electricity Market and the Wholesale Electricity Market.

We also operate retail and wholesale gas markets across south-eastern Australia and Victoria's gas pipeline grid.

Electricity



Gas



Declared Wholesale Gas Market (DWGM)

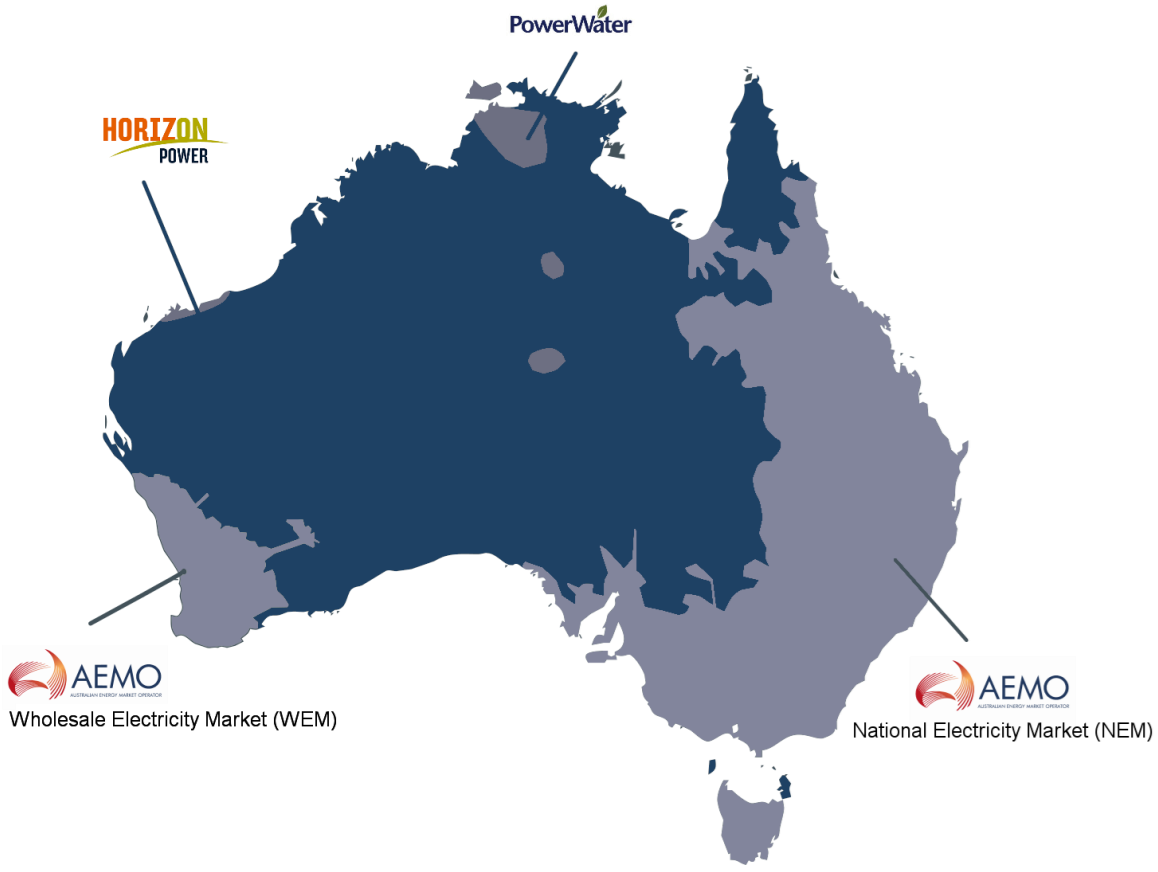


Short Term Trading Market (STTM)

and

Gas Supply Hub (GSH)

Australian Energy Market Operator



We are the independent **energy market and system operator** and **system planner** for the **National Electricity Market (NEM)** and the WA **Wholesale Electricity Market (WEM).**

We also operate **retail and wholesale gas markets across south-eastern Australia** and **Victoria's gas pipeline grid**.

We are a **member-based**, **not-for-profit** organisation.

Expected changes to 2050 (step change scenario)



Storage capacity

to increase by a factor of 30

(Batteries, virtual power plants, pumped hydro.)

NOW	2030	2050
2 GW	15 GW	61 GW



Grid-scale wind and solar

to increase 9-fold

NOW	2030	2050
16 GW	44 GW	141 GW



Distributed solar PV

to increase 5-fold

NOW	2030	2050
15 GW	35 GW	69 GW



Electricity usage from the grid

to nearly double

NOW	2030	2050
180 TWh	184 TWh	320 TWh



Gas-fired peaking plants

to increase

While current mid-merit plants will all retire within that period.

NOW	2050
7 GW	10 GW



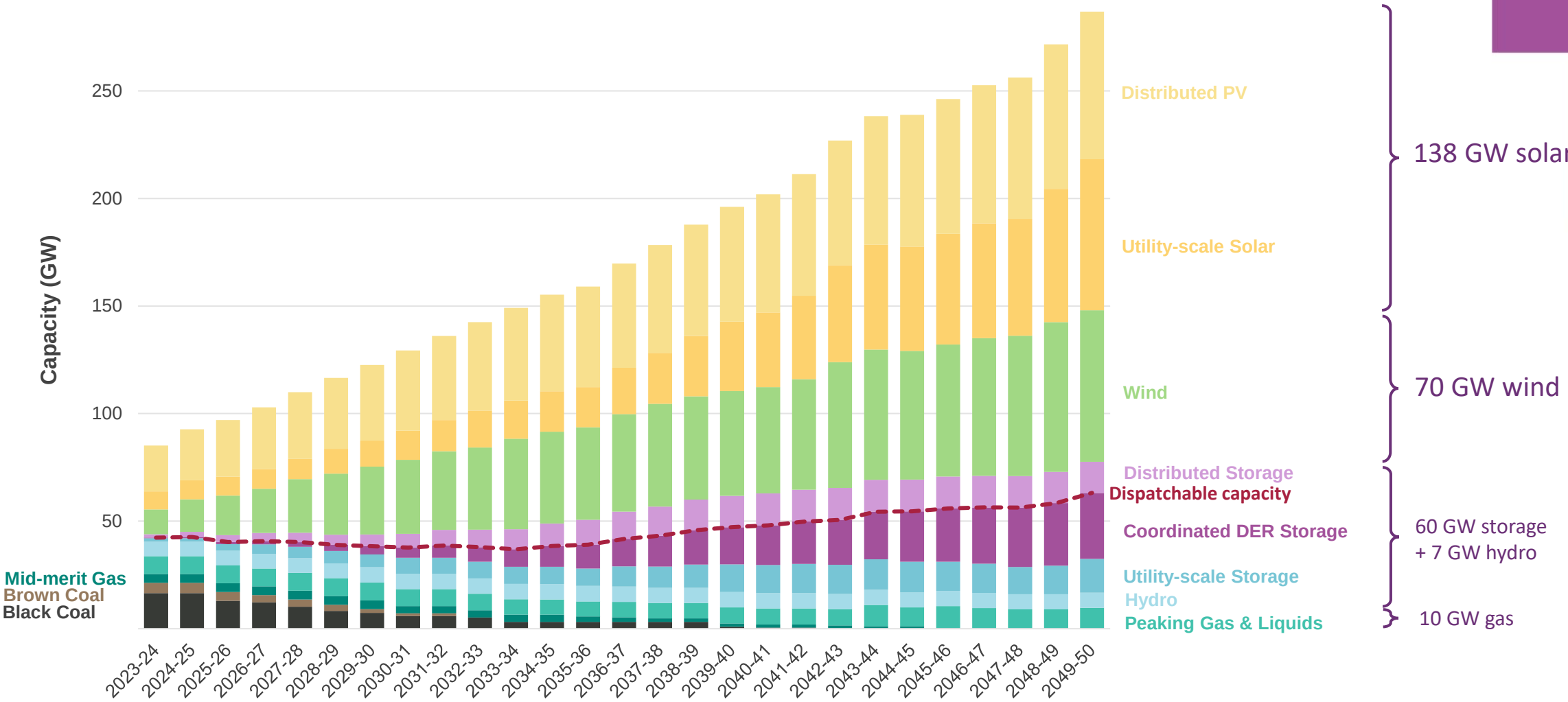
Coal generation

to be withdrawn

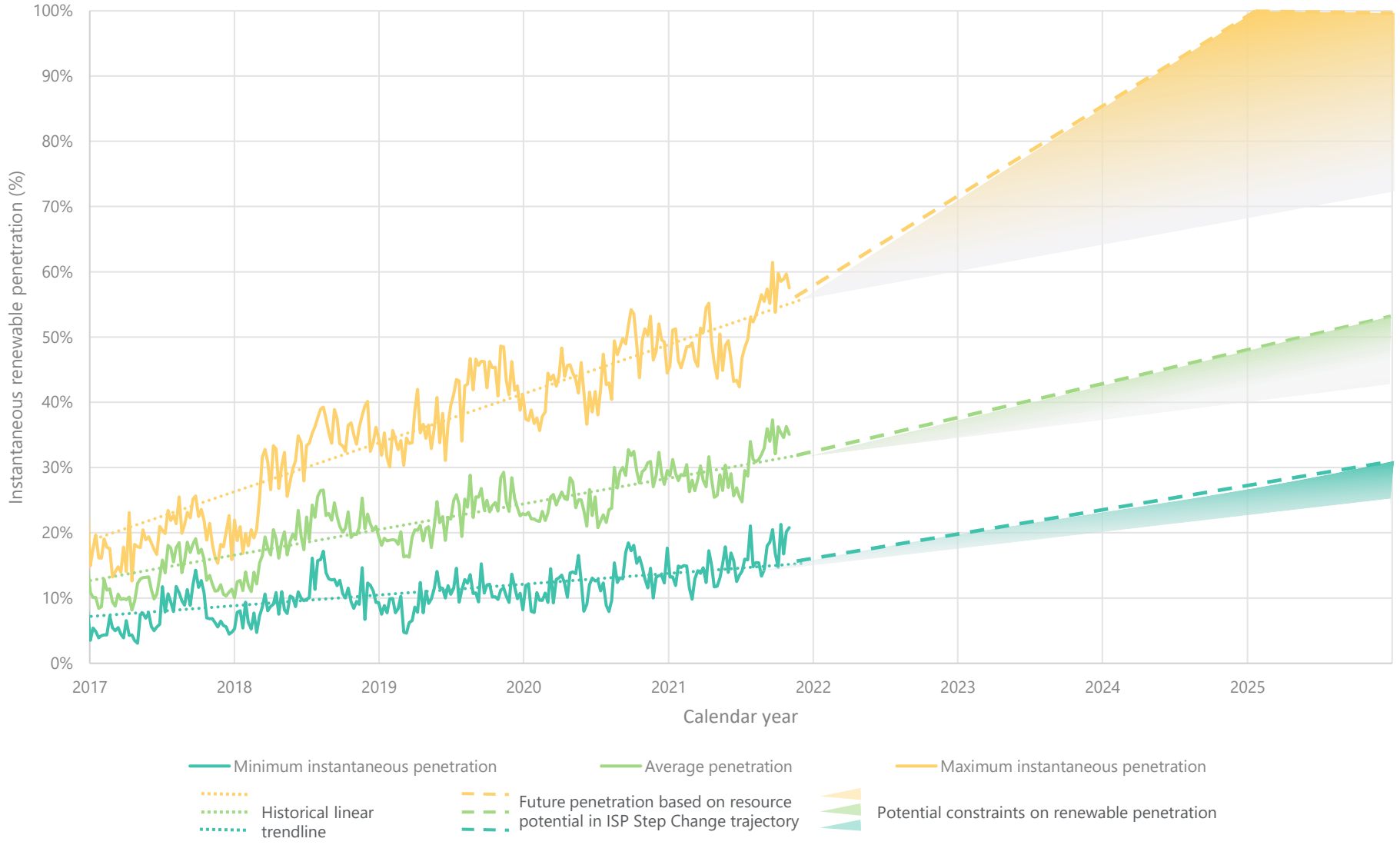
Capacity to be retired by:

2030	2043
60%	100%

Renewable capacity to at least double every decade from now to 2050…



Operational changes in the next five years



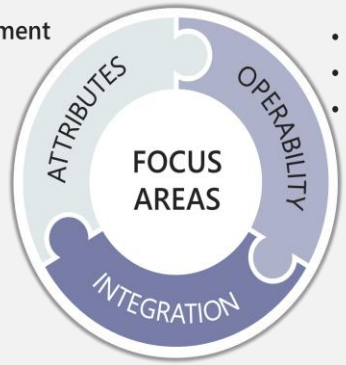
*'Renewable' generation defined as including grid-scale wind and solar, hydro generation, biomass, storage generation and distributed PV (DPV)

Engineering Framework

The Engineering Framework

Current knowledge and work (March 2021)

- Frequency Management
- System Restoration
- System Strength
- Voltage Control
- Resource Adequacy



- System Analysis
- Control Room and Support
- Resilience

- Distributed Energy Resources
- Performance Standards

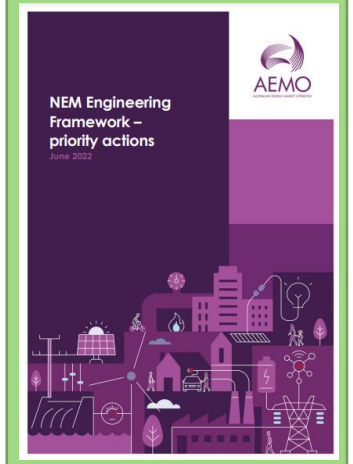
Operational Conditions Summary (July 2021)

1	Fewer synchronous generators online
2	Ubiquitous rooftop solar
3	Extensive grid-scale VRE
4	Structural demand shifts
5	Responsive demand
6	Widespread energy storage

Potential gaps and actions (December 2021)



Priority actions (June 2022)



Roadmap to 100% RE Operation (December 2022)

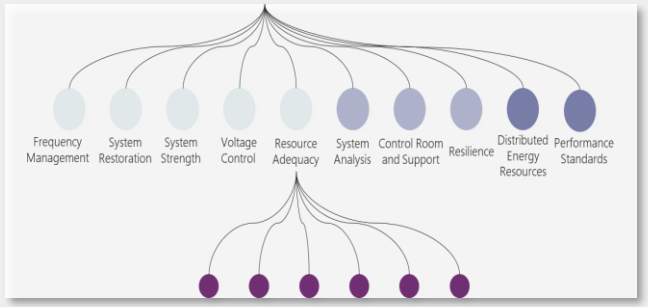


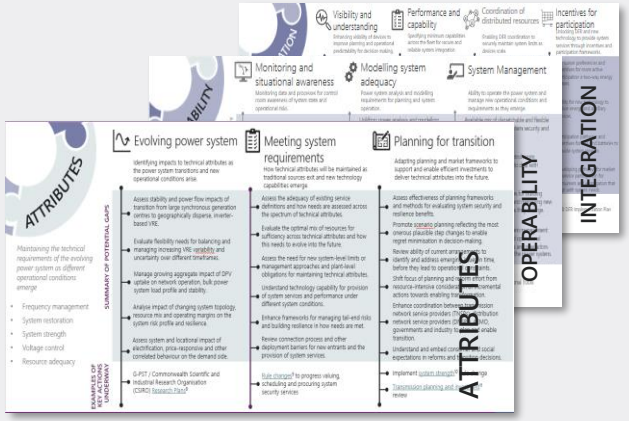
Elements of the Initial Roadmap

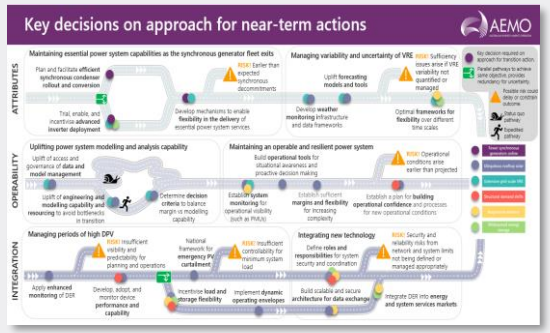
300+ individual potential gaps

Summaries of potential gaps

Key decisions on approach







Prioritisation principles



Value

- Consumer outcomes
- Market benefits
- Power system security

Urgency

- Timing of when issue will arise
- Time taken to complete work
- Timing alignment with committed industry activities

Minimising regrets

- Likelihood of gap occurring regardless of when
- Risk and consequence of being unprepared if gap arises

Leveraging existing processes

- Leveraging existing processes where early action is taken

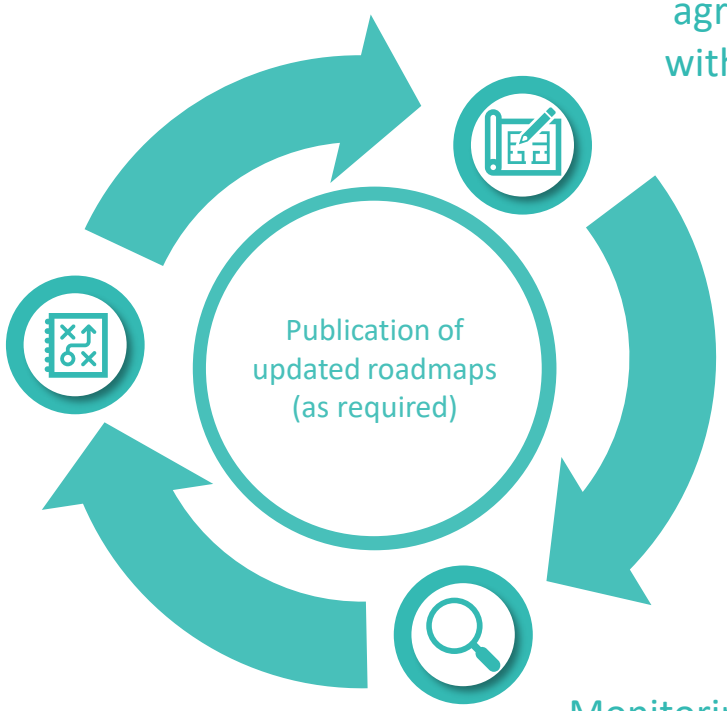
Funding and resourcing

- Suitable funding sources to be confirmed
- Resource implications for all stakeholders

Tackle priorities progressively

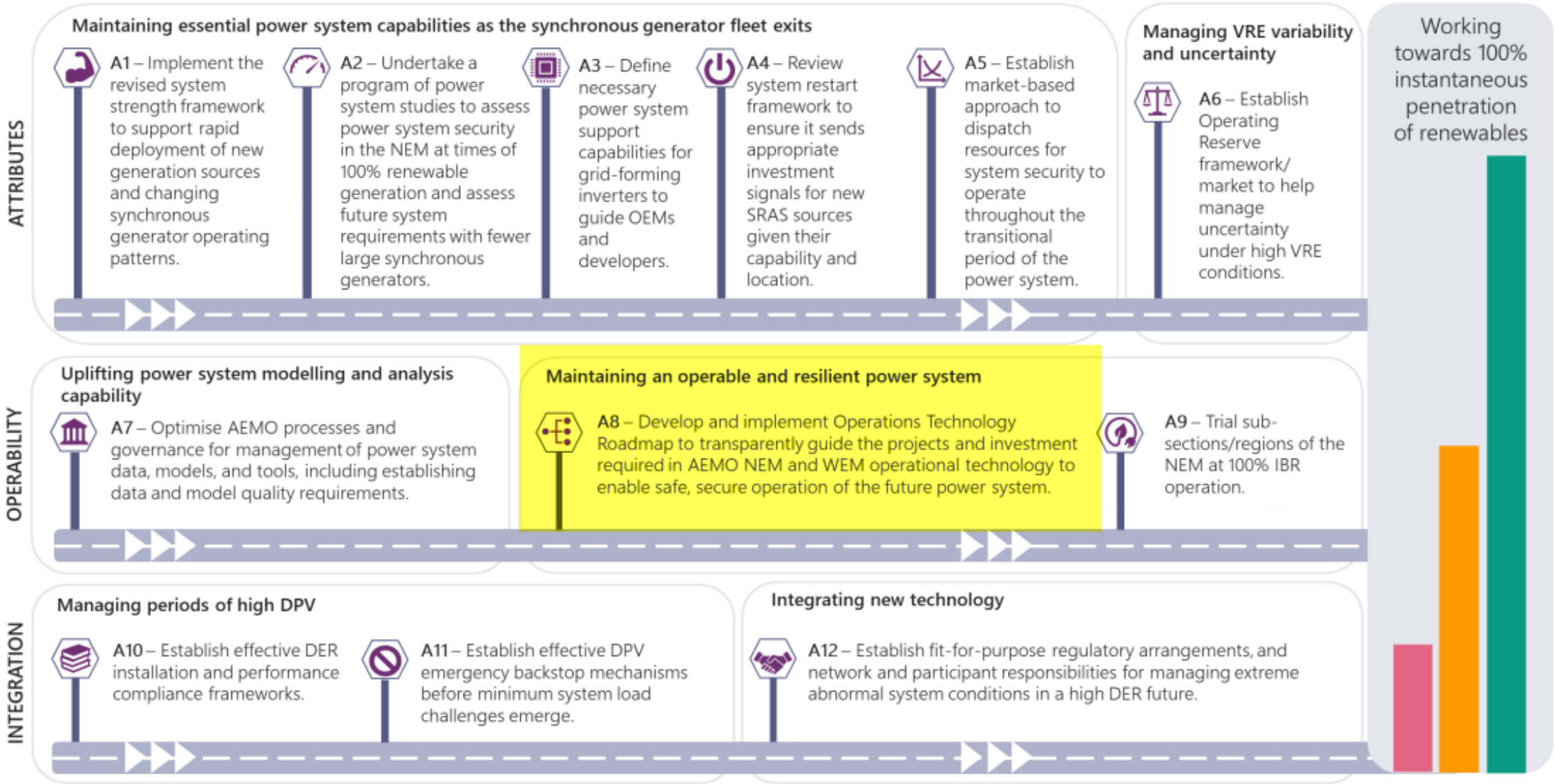
Progressive commencement of agreed priority actions within existing and new implementation workplans.

Prioritise next tranche of gaps, develop actions, and agree responsibilities



Monitoring progress of actions to track industry preparedness for each Operational Condition

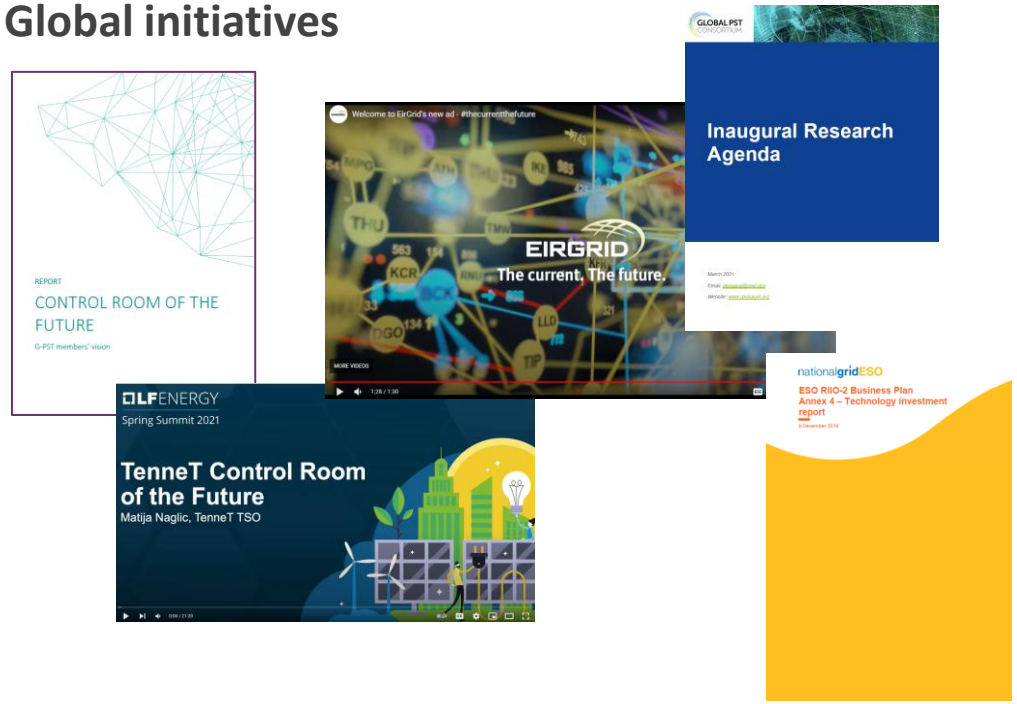
Priorities


ATTRIBUTES
Maintaining essential power system capabilities as the synchronous generator fleet exits
A1 – Implement the revised system strength framework to support rapid deployment of new generation sources and changing synchronous generator operating patterns.
A2 – Undertake a program of power system studies to assess power system security in the NEM at times of 100% renewable generation and assess future system requirements with fewer large synchronous generators.
A3 – Define necessary power system support capabilities for grid-forming inverters to guide OEMs and developers.
A4 – Review system restart framework to ensure it sends appropriate investment signals for new SRAS sources given their capability and location.
A5 – Establish market-based approach to dispatch resources for system security to operate throughout the transitional period of the power system.
Managing VRE variability and uncertainty
A6 – Establish Operating Reserve framework/ market to help manage uncertainty under high VRE conditions.
Working towards 100% instantaneous penetration of renewables
OPERABILITY
Uplifting power system modelling and analysis capability
A7 – Optimise AEMO processes and governance for management of power system data, models, and tools, including establishing data and model quality requirements.
Maintaining an operable and resilient power system
A8 – Develop and implement Operations Technology Roadmap to transparently guide the projects and investment required in AEMO NEM and WEM operational technology to enable safe, secure operation of the future power system.
A9 – Trial sub-sections/regions of the NEM at 100% IBR operation.
INTEGRATION
Managing periods of high DPV
A10 – Establish effective DER installation and performance compliance frameworks.
A11 – Establish effective DPV emergency backstop mechanisms before minimum system load challenges emerge.
Integrating new technology
A12 – Establish fit-for-purpose regulatory arrangements, and network and participant responsibilities for managing extreme abnormal system conditions in a high DER future.

Operations Technology Roadmap

Strategic driver

Global initiatives



CSIRO collaboration



Corporate priority



Engineering framework

A8	Develop and implement Operations Technology Roadmap to transparently guide the projects and investment required in AEMO NEM and WEM operational technology to enable safe, secure operation of the future power system.	Begin implementation of AEMO NEM and WEM Operations Technology Roadmap to enable AEMO to achieve the required uplift in operations technology and capability and help facilitate the renewable energy transition.	Existing AEMO-coordinated action. AEMO Operations Technology Roadmap.	ID053, ID073, ID130, ID412

High level project approach



See the roadmap here:

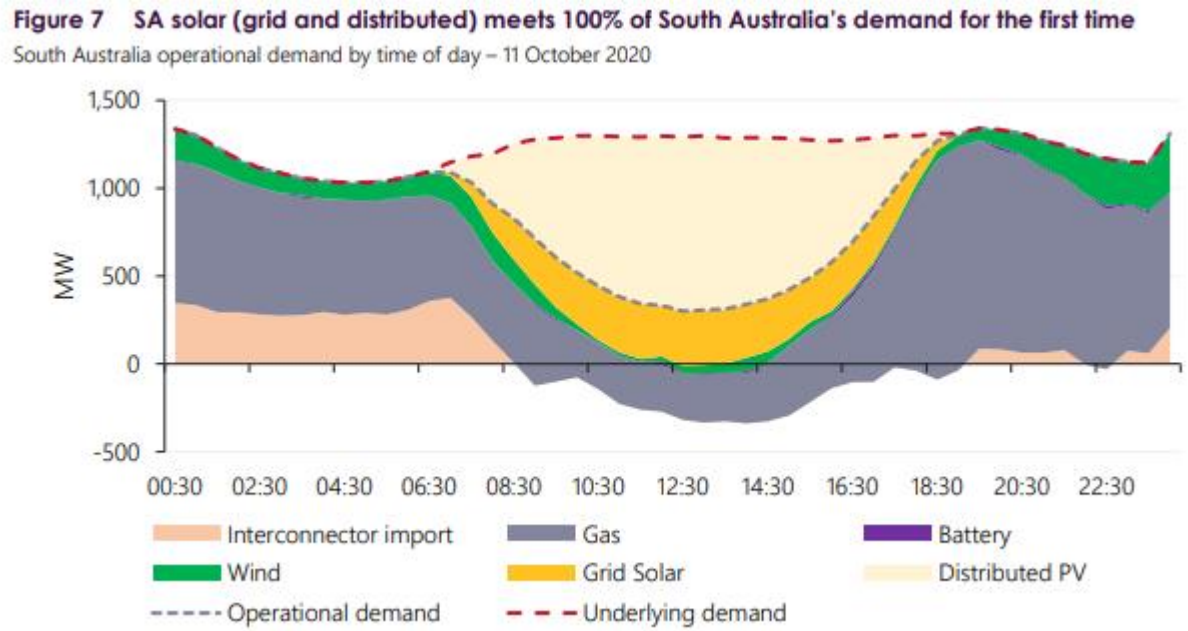
https://aemo.com.au/initiatives/major-programs/operations-technology-roadmap

Emerging operational risks

- **Management of system strength**



- **Ramping**



https://aemo.com.au/-/media/files/major-publications/qed/2020/qed-q4-2020.pdf?la=en

- **Market interventions**

Individual tool roadmaps

Why



The Current State and Future Vision



How the Operational Need is Changing – EF Operational Conditions



Key Drivers for Change (EF and OTR Gaps) and Actions

What



Business Capability Model Nodes



Data Requirements



OT Tool Requirements

How



Risk Assessment



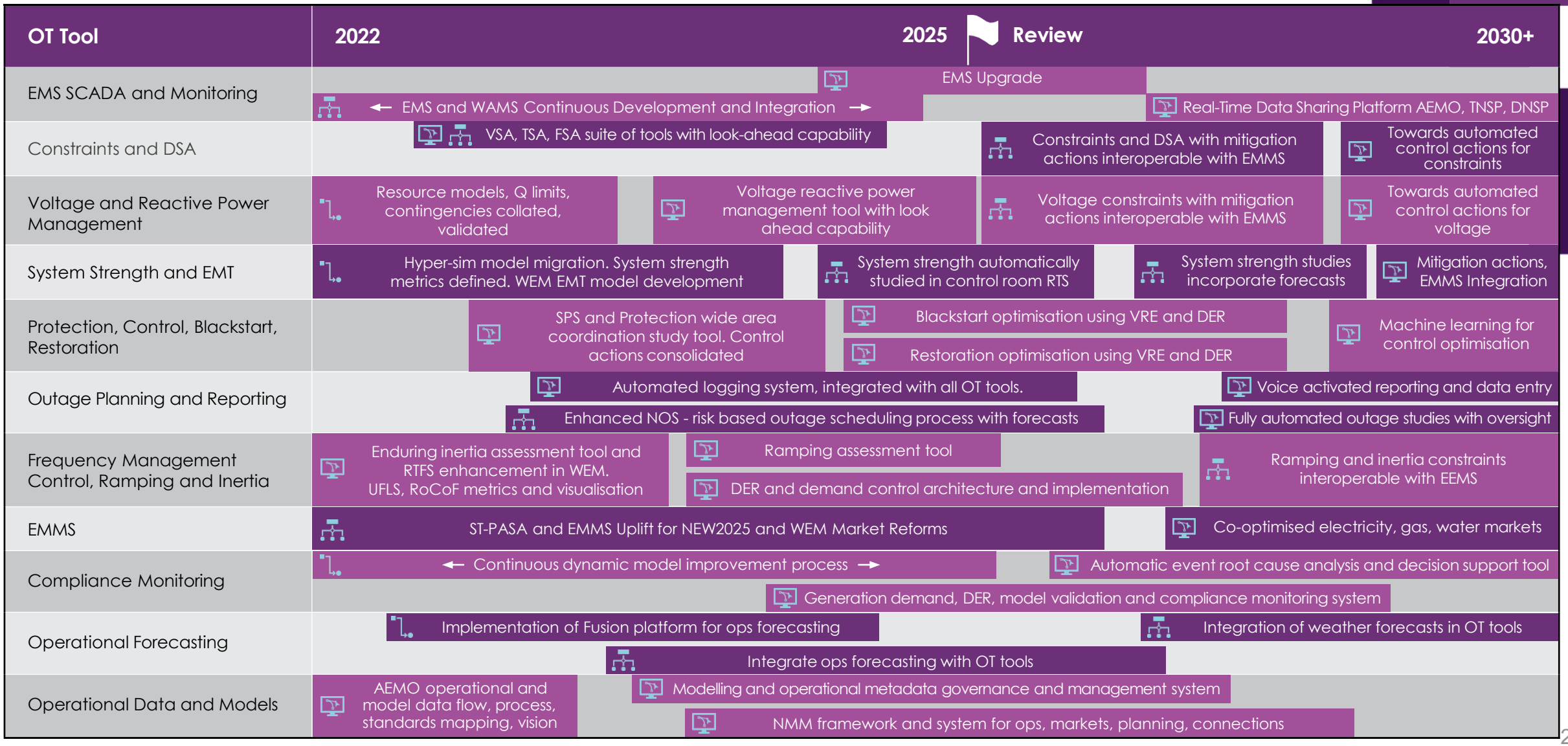
Options and Estimated Cost Benefit



A Roadmap for Development through 2030+

Operations Technology Roadmap

Key: New Tool/Process Enhanced | Enhanced Existing Tool | Existing Process/Project



Note: A number of tools will be introduced into WEM operations as part of WEM reform in the coming years.

Further reading

Engineering Framework

https://aemo.com.au/initiatives/major-programs/engineering-framework

Energy Systems Integration Group

https://www.esig.energy/

Operations Technology Roadmap

https://aemo.com.au/initiatives/major-programs/operations-technology-roadmap

Global Power System Transformation Consortium (GPST)

https://globalpst.org/

AEMO

AUSTRALIAN ENERGY MARKET OPERATOR

For more information visit

aemo.com.au